

FRISBEE

Presentation Speaker Notes

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github.com/bhavishyasr/frisbee

How to use these notes. Each section maps to one slide. The *Say* points are things you can say almost verbatim — keep them conversational, not read aloud. *Key terms* remind you of what is on screen. *Tip* notes flag common reviewer questions or pacing advice. Total target time: **12–15 minutes** across 14 slides (≈ 60 seconds per content slide).

1

SLIDE FRISBEE — Personal Behavioral Alignment Modeling

Who speaks: Bhavishya (or whoever opens)

Say:

- *“Good morning. We are Bhavishya, Arya, Hansika, and Manogna from the Department of AI and Machine Learning. We are here to present FRISBEE — Functional Realtime Intelligence for Spotting Blissful Everyday Actions.”*
- *“The name is a backronym, but the idea underneath it is serious: we built a machine learning system that learns your behavioral patterns from your own words — and uses them to predict your week better than you predict it yourself.”*
- *“Our GitHub repository is open — the link is on screen. The entire system is running and functional.”*

Tip: Keep this under 30 seconds. The title slide is just an introduction — the energy should feel confident, not rushed. Make eye contact, do not read from a screen.

1

SLIDE What FRISBEE is, in one paragraph

Who speaks: Arya (or second presenter)

Say:

- *“FRISBEE is a personal behavioral modeling system. Every day you write a short free-form message — just how your day went, in your own words. Over time, a logistic regression model trained only on your messages learns to predict whether your week will be a yes-week or a no-week.”*
- *“The model works with four handcrafted semantic clusters — **drift, align, excuse, and***

energy — which become a four-dimensional TF-IDF feature vector. That vector goes into a binary classifier.”

- “The core hypothesis is this: a model trained on your own language will outperform your own blinded prediction of whether your week went well — measured by mean absolute error over a held-out test period.”
- “And critically — all computation happens on your device. No server. No cloud. No data leaves your machine.”

Tip: If a reviewer asks “Why logistic regression?” — answer: “Because we have four features and a binary output. Logistic regression is the correct, interpretable, and mathematically honest choice for this problem. We do not need a neural network for $\mathbb{R}^4 \rightarrow \{0,1\}$.”

1

SLIDE The Identity Gap

Who speaks: Hansika

Say:

- “The problem we are solving is not a productivity problem. It is a self-knowledge problem. There is a consistent, measurable gap between who people believe themselves to be and who actually shows up day to day.”
- “Teenagers feel this most acutely. “I said I would study. I did not.” “Why did I do that again?” This is not laziness — it is optimism bias. Morning-you makes promises that evening-you may not keep.”
- “The ML question is: can a model learn your specific drift patterns from your own words, better than you can predict them yourself? That is what FRISBEE answers.”

Tip: This slide sets the emotional and conceptual foundation. Speak a little slower here. Let the framing land before moving on. The three column cards on screen (Optimism Bias, Not a Tracker, The ML Question) give you visual anchors — point to each one as you mention it.

1

SLIDE Four numbered goals

Who speaks: Manogna

Say:

- “Our objective has four parts. First, we build a personal logistic regression model trained solely on one individual’s messages — no one else’s data enters the model.”
- “Second, we predict a binary weekly alignment label from a four-cluster TF-IDF feature

vector.”

- *“Third, we show that the model’s predictions converge toward ground truth faster than the user’s own self-predictions. We measure this with MAE.”*
- *“Fourth, this is individual-level optimism bias modeling. Not population statistics. Your bias. Your model. Nobody else’s.”*

Tip: Walk through the four points numbered on screen. You can say *“First... Second... Third... Fourth...”* and point briefly. This slide is fast — about 45 seconds.

1

SLIDE In scope / Out of scope / Stack

Who speaks: Bhavishya

Say:

- *“We were disciplined about scope. In scope: single user, on device, free-form text, weekly binary label, and four fixed semantic clusters with blended IDF for cold start.”*
- *“Out of scope: multi-user models, automated labeling, real-time prediction, and any external ML library. This is a clean boundary — we did not overpromise.”*
- *“The implementation is TypeScript at 96 percent of the codebase. React 19, Vite 7, Tailwind CSS. The ML engine is pure TypeScript — no TensorFlow, no scikit-learn, no external dependencies of any kind. Everything runs in the browser, persisted in IndexedDB.”*

Tip: Reviewers often appreciate honest scope limitation. If asked *“why not multi-user?”* — answer: *“The model is non-transferable by design. Weights learned on one person’s language carry no valid signal about another person. Multi-user is a future direction, not an oversight.”*

1

SLIDE Six academic references

Who speaks: Arya

Say:

- *“Our literature survey grounds the project in established cognitive science and trust research.”*
- *“Sweller’s Cognitive Load Theory tells us that when mental capacity is depleted, people default to low-effort shortcuts. This explains why knowing better does not mean doing better.”*
- *“Kahneman’s System 1 and System 2 framework reinforces this — fatigue degrades deliberate thinking, producing habitual poor choices. Rebetez and Chwolka and Rott extend this*

into procrastination and decision deferral specifically.”

- *“Lee and See, and Hengstler give us our trust framework. The model should neither be over-trusted nor ignored. Trust must be calibrated to actual reliability — which is exactly what our convergence metric tracks.”*

Tip: You do not need to explain every paper in depth. Name each one with one sentence and move on. The six references are on screen — let the slide do the work. This slide should take about 60–75 seconds.

1

SLIDE Four categories of prior work and their gap

Who speaks: Hansika

Say:

- *“We looked at four categories of existing tools. Habit trackers like Streaks and Habitica count completions — they do not analyze language, they do not predict, they do not personalize behavioral signal.”*
- *“Journaling apps like Day One store text but run no modeling. Any insight they offer is surface sentiment, not behavioral prediction.”*
- *“Population NLP APIs train on crowd data and apply the model to you. Your individual patterns become noise — averaged away.”*
- *“Generic recommenders suggest behavioral changes but have no model of the gap between your intention and your action.”*
- *“The common gap: all of them are built for populations, not persons. None of them model the intention-to-action gap from the user’s own language. That is exactly what FRISBEE does.”*

Tip: The amber alert block at the bottom of the slide is the key takeaway. Point to it when you say the last line. If asked *“what about GPT-based journaling?”* — *“Population-trained models applied to a single user still face the same averaging problem. FRISBEE’s model is trained exclusively on you.”*

1

SLIDE Six-step pipeline

Who speaks: Manogna

Say:

- *“The methodology is a six-step loop that runs every day and closes weekly.”*
- *“Step one: you write a free-form message. Step two: the tokenizer in `tokenize.ts` pro-*

cesses it — it is apostrophe-preserving, so contractions like “didn’t” are kept intact and not broken apart.”

- “Step three: *features.ts* scores the message against the four clusters using TF-IDF, producing a vector in $\mathbb{R}^4$.”
- “Step four: that vector goes through the logistic regression model — $z = w^T x + b$, then $p = \sigma(z)$ — giving a probability that this will be a yes-week.”
- “Step five: at the end of the week, you label it — yes or no. That is the ground truth. Step six: the model updates its weights using L2-regularized gradient descent on those weights only.”
- “All of this runs in *engine.ts*, persisted in *vault.ts* via IndexedDB. Zero data leaves your device.”

Tip: Walk the six numbered points on screen one by one. The math is simple enough to say aloud — do not skip it, but do not dwell on it either. This slide typically draws the most questions.

1

SLIDE Three takeaways (dark green slide)

Who speaks: Bhavishya

Say:

- “We place the conclusion here, before the technical deep-dive, so the committee has the “so what” before we go into the how.”
- “Three things FRISBEE demonstrates. First: personal language is predictive. Your daily words encode your behavioral alignment signal — drift, alignment, excuses, energy.”
- “Second: individual beats population. A model trained only on your data outperforms one trained on anyone else’s.”
- “Third: optimism bias is modelable. The gap between morning-you and evening-you is consistent enough to represent in four dimensions.”
- “The whole machine is personal, private, on-device, and honest. That is the design principle as much as the technical spec.”

Tip: This slide has a visual impact — white text on dark green. Pause briefly when it appears. Speak these three points slowly. They are the thesis of the project.

1

SLIDE The architecture diagram (full slide)

Who speaks: Arya

Say:

- *“This is the full system architecture. Let me walk you through it top to bottom.”*
- *“At the top: the user inputs a free-form daily message. That flows as a token stream into `tokenize.ts`, then into `features.ts` which produces the $\mathbb{R}^4$ TF-IDF vector.”*
- *“That vector enters `model.ts` — logistic regression — which produces a probability. At the end of the week, the user provides a binary label. The update is computed and the model adapts.”*
- *“On the right side is the Event Bus in `bus.ts` — a typed pub-sub system that decouples the mascot, haptics, and UI from the core ML engine. It uses discriminated union events in TypeScript.”*
- *“At the bottom left, `scientist.ts` reads from the engine and surfaces word-delta hypotheses — it tells you which words are shifting your predictions.”*
- *“Everything is persisted in `vault.ts` via IndexedDB. No backend. No API. No cloud.”*

Tip: Use a pointer or gesture toward the diagram. Do not just describe it — *navigate* it. Committee members will follow along visually. This slide can take up to 90 seconds.

1

SLIDE Feature engineering and logistic regression math

Who speaks: Hansika

Say:

- *“Step one is feature engineering. We define four clusters: drift, align, excuse, and energy. Each cluster is a hand-tuned list of words — for example, drift contains words like “procrastinated”, “wasted”, “scrolled”. Align contains “focused”, “proud”, “accomplished”.”*
- *“We score each message against these clusters using TF-IDF. Rare-but-used words carry more signal than frequent ones. To handle cold start, we blend personal IDF with background IDF — 40 percent personal, 60 percent background — so the model is not completely blind in the first few weeks.”*
- *“Step two is the model itself. The feature vector is $x = [x_{\text{drift}}, x_{\text{align}}, x_{\text{excuse}}, x_{\text{energy}}]^\top$. We compute the linear combination $z = w^T x + b$, pass it through a sigmoid to get p , and compute binary cross-entropy loss.”*
- *“The weight update applies L2 regularization to w only — not the bias. $\eta = 0.1$, $\lambda = 0.01$. These hyperparameters were chosen to give stable convergence over a short data series.”*

Tip: If asked “*why not use a neural network?*” — “Four features. Binary output. Weekly updates on a tiny dataset. Logistic regression is the right tool. A neural network would

overfit immediately and be completely uninterpretable.”

If asked about the blended IDF split: *“0.4/0.6 was calibrated empirically. The background corpus prevents complete randomness at cold start while still allowing personal vocabulary to dominate over time.”*

1

SLIDE Four evaluation dimensions

Who speaks: Manogna

Say:

- *“We evaluate on four dimensions. The primary metric is mean absolute error. We compare the model’s MAE against the user’s own blinded self-prediction MAE over the held-out test period. That is the direct test of our hypothesis.”*
- *“We also track binary accuracy — what fraction of weeks did the model’s yes-or-no call match the ground truth label.”*
- *“Calibration asks whether a predicted probability of 0.7 is actually correct about 70 percent of the time. We use a reliability diagram for this.”*
- *“Convergence tracks the norm of the weight update per cycle. This tells us how many weeks of data the model needs before it stabilizes.”*
- *“One honest limitation: our held-out test is four weeks. That is four data points. We are explicit that this is a proof of concept, not a statistically generalisable result.”*

Tip: Reviewers will almost certainly ask about the sample size. Preempt it by saying the last bullet point yourself — it shows honesty and academic rigour, not weakness.

1

SLIDE Six acknowledged limitations

Who speaks: Bhavishya

Say:

- *“We are transparent about our constraints. Sample size: four weeks of held-out data means four data points. We cannot claim statistical significance. We do not.”*
- *“Cold start: the first two or three weeks produce near-random predictions. Blended IDF reduces but does not eliminate this.”*
- *“Feature space: our four clusters are hand-tuned. A user who habitually uses none of those words will not be well-served. There is no out-of-vocabulary expansion in the current version.”*
- *“Self-report bias: the ground truth label y is user-provided. How you feel when you label*

the week affects what signal the model receives. This is an inherent limitation of any self-report system.”

- *“The model is non-transferable. It is single-user by design. And from Lee and See and Hengstler, we know trust calibration is delicate — early errors risk under-trust, overconfident predictions risk over-reliance.”*

Tip: Presenting constraints confidently is a sign of academic maturity. Do not apologise for them — explain them. The committee should see that you understand the boundary of your own claims.

1

SLIDE Six planned extensions

Who speaks: Arya (or rotate back to opener for symmetry)

Say:

- *“We have six clear directions for future work.”*
- *“First, extending the held-out test period beyond four weeks — this is the most immediate step toward statistical validity.”*
- *“Second, dynamic vocabulary — letting the model discover user-specific signal words outside the fixed four clusters, using online feature selection.”*
- *“Third, temporal decay — weighting recent messages more heavily, since behavioral patterns shift over months and old data should have diminishing influence.”*
- *“Fourth, showing the model’s confidence to the user, per Hengstler 2020 — so trust is calibrated, not blind.”*
- *“Fifth, a multi-user study to enable between-person comparison of optimism bias magnitude.”*
- *“Sixth, using message length and complexity as a proxy for cognitive load — Sweller’s framework applied in real time to predict a bad decision before it happens.”*
- *“Thank you. We are happy to take questions.”*

Tip: End on a confident note, not a trailing voice. *“Thank you. We are happy to take questions.”* Make eye contact with the committee when you say it. Do not add *“that’s it”* or *“so yeah.”*

1

1. Why logistic regression and not something more sophisticated?

Four features, binary output, weekly updates, tiny dataset. Logistic regression is the correct, interpretable, and statistically honest model for this problem. A neural network would overfit immediately and produce uninterpretable weights.

2. The dataset is only four data points. How can you claim anything?

We do not claim statistical generalisability. We claim proof-of-concept — that the pipeline is functional and the hypothesis is testable. Statistical validity is the first future enhancement. Honesty about sample size is a feature, not a bug.

3. What if a user does not use any words in your four clusters?

The blended IDF and background corpus provide a floor. However, this is a genuine limitation acknowledged in Constraints. Dynamic vocabulary expansion is the direct fix — listed as Future Enhancement 2.

4. How is this different from journaling apps with sentiment analysis?

Journaling apps run population-trained models on your text. FRISBEE's model is trained exclusively on your own messages. Your patterns are signal, not noise. The prediction target is also different: not sentiment, but behavioral alignment as *you* define it.

5. Is the ground truth reliable if users label it themselves?

No, and we say so explicitly in the Constraints slide under Self-Report Bias. Recall bias and mood at labeling time both introduce noise. A future enhancement could explore more objective behavioral proxies.

6. Could this work on a multi-user level?

Not with the current architecture — weights are personal and non-transferable. A multi-user study is listed as Future Enhancement 5, which would aggregate *outcomes* across users, not share model weights.

7. What is the Event Bus and why does it exist?

The Event Bus in `bus.ts` decouples the mascot, haptics, and UI feedback from the ML engine using a typed pub-sub system in TypeScript. It keeps the core ML layer pure — no UI dependencies inside the learning logic.